

AROOJ FATIMA

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EDUCATION

National University of Sciences & Technology (NUST)

Sep 2023–Jun 2027

Bachelor of Science in Bioinformatics, Islamabad, Pakistan

CGPA: 3.41/4.00

Relevant Coursework: Discrete Mathematics, Data Structures and Algorithms, Design and Analysis of Algorithms, Operating Systems, Computer Organization and Assembly Language, Parallel & Distributed Computing, Linear Algebra, Probability and Statistics, Software Engineering, Fundamental of Computer Programming, Object Oriented Programming, Database Systems

RESEARCH INTERESTS

Formal Verification, Interactive Theorem Proving, Systems Software Verification, C Program Verification, Programming Languages and Formal Semantics, and Verified Compilation.

RESEARCH & PROFESSIONAL EXPERIENCE

Formal Verification Engineer Intern

Jun 2026–Present

Formal Land, Paris, France / Remote

[rocq-of-rust](#)

- Updated rocq-of-rust for a newer version of Revm (a Rust implementation of the Ethereum Virtual Machine), adapting its Rust-to-Rocq translations, functional specifications, and proofs to changes in Revm’s architecture, and added evaluation through OCaml extraction.
- Currently develop instruction-level correspondence proofs between Revm and go-ethereum in Rocq, relating their models of stack, memory, storage, gas, host state, control flow, and errors. Also built mutation tests and runners for official Ethereum test cases.

Student Researcher

Jun 2025–Present

System Analysis and Verification Lab, NUST

[Playground](#)[Algorithms](#)

- Built a foundation in interactive theorem proving through Lean 4 game-based exercises and small formalizations. I then moved to Rocq because of its mature ecosystem (C verification libraries) for verifying systems.
- Proved correctness properties of BST insertion, insertion sort, merging sorted lists, and MPI block-decomposition formulas, and formalized red-black-tree invariants and repair operations.
- Currently working on the broader formal verification of the $\mu\text{C}/\text{OS-III}$ real-time kernel using Rocq and the Verified Software Toolchain.

Summer Research Intern

Jun–Aug 2025

Deep Learning Laboratory, NCAI, NUST

[Repository](#)[Certificate](#)

- Training models across decision trees, neural networks, ResNet50, and YOLO on CIFAR and custom datasets
- Reproduced active-learning research for medical-image analysis and compared selection methods using TF-IDF and Sentence-BERT representations.

ACCEPTED RESEARCH

- [1] **Arooj Fatima and Saad Saghir.** “Towards Machine-Checked Verification of the $\mu\text{C}/\text{OS-III}$ Kernel Using Rocq and the Verified Software Toolchain.” *2026*

Verified with VST four C priority-bitmap functions for a 32-priority $\mu\text{C}/\text{OS-III}$ configuration: initialization, insertion, removal, and highest-ready-priority lookup. Proved that their bit-level behavior satisfies abstract ready-priority specifications.

Accepted for presentation at the **FMCAD 2026 Student Forum.**

[Repository](#)

- [2] **Arooj Fatima, Saad Saghir and Adnan Rashid.** “A Comparative Formal Verification Framework for General-Purpose CPU Scheduling in Rocq.” *2026*

Developed Rocq models and machine-checked proofs for FIFO, SJF, Lottery, MLFQ, Stride, and CFS, including a red-black-tree-based CFS variant. Identified reusable minimum-key, layered-priority, and ticket-interval proof patterns.

Accepted for publication at the **6th IEEE International Conference on Digital Futures and Transformative Technologies (ICoDT2 2026).**

[Repository](#)

OPEN-SOURCE CONTRIBUTIONS

Contributor to coqutil and Parseque	<i>Jul 2026–Present</i>
coqutil and Parseque	Rocq
- Authored six merged pull requests adding build documentation, reusable lemmas for lists and optional values, and multiline source-position tests. - Contributions: #168, #169, #170, #171, #176, #178, and #22.	

PROJECTS & SELF-LEARNING

Software Foundations	<i>Dec 2025–Present</i>
<i>Rocq</i>	Private repository
Working through the Software Foundations in Rocq and completing the chapter exercises. The repository is private because the book asks readers not to post solutions publicly.	

Verified Distributed Protein-Network Analysis	<i>May 2026</i>
<i>PySpark, Rocq, and STRING Part of Parallel & Distributed Computing Course</i>	Repository
- Built a PySpark pipeline that processed 13.7 million protein interactions into 473,860 high-confidence edges over 16,201 proteins. - Formalized the Large-Star and Small-Star connected-components transformations in Rocq and proved their correctness at a fixpoint.	

Concurrent Web-Server Simulation on FreeRTOS	<i>Dec 2025</i>
<i>C, FreeRTOS, RISC-V, and QEMU Part of Operating Systems Course</i>	Repository
Built a concurrent web-server simulation demonstrating priority-based scheduling, producer-consumer task queues, and mutex-based synchronization for HTTP requests.	

Parallel Algorithm Performance Analysis	<i>Nov 2025</i>
<i>C, OpenMP, Pthreads, and MPI Part of Operating Systems Course</i>	Repository
Implemented and benchmarked parallel matrix multiplication and merge sort on datasets ranging from 10 to 10 million elements, comparing the overhead and scaling behavior of three parallel-programming models.	

Operating Systems Concepts OSTEP	<i>Nov 2025</i>
<i>C</i>	Repository
Implemented core Operating Systems concepts from OSTEP textbook including process API (fork/exec/wait), CPU scheduling algorithms, and memory virtualization through hands-on coding exercises and simulations.	

OTHER ACTIVITIES

Curriculum Design and Mentorship Team	<i>Oct 2025–Present</i>
<i>SPARK</i>	
Research Opportunities List Curator	<i>Dec 2025–Mar 2026</i>
<i>Independent GitHub repository</i>	Repository
Technical Notes and Blogging	<i>Jan 2025– Nov 2025</i>
<i>Writings on theoretical CS, formal verification, quantum computing, and operating systems</i>	Notion Medium
Executive, Admin Events	<i>Oct 2024–Apr 2025</i>
<i>NUST Entrepreneurs Club</i>	Certificate
Deputy Director, Admin Events	<i>Oct 2024–Apr 2025</i>
<i>NUST IEEE Society</i>	
Organizing Committee Volunteer	<i>Oct 2024</i>
<i>ICoDT2 2024 Conference</i>	Certificate

AWARD

FMCAD 2026 Student Forum Travel Award for presenting report at TU Graz, Austria	<i>2026</i>
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TECHNICAL SKILLS

Verification: Rocq, Lean 4, Verified Software Toolchain
Programming: Rust, C, OCaml, Python
Systems and Tools: Git, Linux, FreeRTOS, MPI, OpenMP, L^AT_EX